Individual Contribution Report

Distributed Cuckoo Filter on ESP32 over ESP-NOW

PDC, Semester 6, Spring 2026 | Group 08 | Institute of Business Administration, Karachi

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Overview

The project was completed collaboratively, with significant portions of the work carried out jointly during lab sessions and coordinated remotely. The sections below reflect each member's primary area of responsibility. Some degree of overlap was inherent given the integrated nature of the system.

Ayesha Kashif

Student ID: 29155

Primary area: Testing, Master Node Logic, Integration, Performance Evaluation

Ayesha was responsible for the master node firmware, which constitutes the coordination layer of the system. Her work encompassed the hash-based routing logic that maps incoming operation requests to the appropriate slave node, the stop-and-wait retransmission loop that compensates for the lack of built-in acknowledgment in ESP-NOW, and the serial command-line interface through which the cluster is operated during testing and demonstration.

She designed and maintained the complete test suite, covering all four suites: filter unit tests, end-to-end cluster simulation, protocol layout verification, and the FPR benchmark. A particularly significant contribution was establishing the native desktop build configuration that allows the full test suite to compile and execute without physical hardware, substantially reducing the time required for iterative development.

Ayesha also conducted the hardware benchmark sessions on the physical ESP32 boards, measuring end-to-end operation latency and sustained throughput, and produced the performance evaluation reported in the final paper. She was additionally responsible for system integration, ensuring that the individually developed modules composed correctly into a functioning distributed system.

Syed Hamza Ahsan

Student ID: 28903

Primary area: Cuckoo Filter Implementation

Hamza was responsible for the implementation of the core data structure: the single-node Cuckoo Filter. This work included the fingerprint derivation scheme, the dual-bucket insertion and lookup logic, the displacement loop bounded at 500 kickout steps, and the single-entry victim cache that maintains filter consistency when the displacement chain exhausts its limit without locating a free slot.

He implemented the alternate-bucket formula adapted for a power-of-two table size and verified that the resulting false positive rate at the chosen configuration (256 buckets, 4 slots per bucket, 8-bit fingerprints) matched the theoretical prediction of approximately 3 percent. The cuckoo_filter module was written with no platform dependency, enabling compilation and execution on a desktop environment independently of any embedded toolchain.

Hamza served as the primary reference for questions regarding filter correctness throughout the project. He also assisted in diagnosing edge cases in the cross-partition kickout chain protocol that surfaced during integration and end-to-end testing.

Kabeer Jafri

Student ID: 29027

Primary area: Communication Layer, WebSocket Implementation, PlatformIO Setup

Kabeer established the PlatformIO project structure at the outset of the project, configuring the build environments for the master firmware, the slave firmware, and the native desktop test target. This setup enabled the multi-environment build workflow used throughout all three milestones and ensured that the codebase remained buildable across both embedded and desktop targets without modification.

He implemented the WebSocket-over-TCP transport layer used in Milestones 1 and 2, which included the message framing logic, connection lifecycle management on the master side, and the request handler infrastructure on each slave. The framing overhead and per-connection heap pressure he documented during this phase directly informed the rationale for migrating to ESP-NOW in Milestone 3.

Kabeer remained involved during the transport migration, contributing to validation testing that compared the behavior of the ESP-NOW implementation against the prior WebSocket baseline. He also assisted with the Wokwi simulation configuration used to verify multi-slave behavior beyond the two-board limit of the physical test environment.

Qazi Muhammad Waiz

Student ID: 28982

Primary area: ESP32 Hardware Setup and Connectivity

Waiz was responsible for all aspects of physical hardware preparation and connectivity. His work included reading MAC addresses from each ESP32 board, maintaining the config.h peer address table, and managing the serial port permissions and firmware flashing procedure on Linux, which required ongoing attention throughout the hardware testing phase.

He handled the ESP-NOW peer registration configuration and was responsible for diagnosing a category of silent delivery failure that occurred when incorrect MAC addresses were present in the configuration. In this failure mode, the master-side send call returns success at the MAC layer while the slave receives nothing, making the fault non-obvious to detect without direct hardware observation.

Waiz established the two-board physical test environment and executed the standard functional verification sequences (insert, lookup, delete, lookup) across multiple sessions to confirm consistent hardware behavior. He also configured and monitored the heartbeat reporting mechanism, verifying that the master node correctly registered each slave as online within the expected time window following power-on.

Summary

Name	ID	Primary Responsibility
Ayesha Kashif	29155	Testing, master node, integration, performance evaluation
Syed Hamza Ahsan	28903	Cuckoo filter implementation
Kabeer Jafri	29027	WebSocket transport, PlatformIO build setup
Qazi Muhammad Waiz	28982	ESP32 hardware setup and connectivity